

Table 1. RealToxicityPrompts results with evaluated-model KV cache disabled. Repeated rows report mean $\pm$ standard error; compact single-pass rows report means without an error term. TBD marks unrun model-method pairs.

	Method	Toxic continuations (%) $\downarrow$	Distinct-2 $\uparrow$	Perplexity $\downarrow$	Safe-concept relevance (0-2) $\uparrow$	Instruction relevance (0-2) $\uparrow$	Fluency (0-2) $\uparrow$	Overall steering (0-2) $\uparrow$
Gemma-2-2B	Original	3.80 $\pm$ 0.26	0.657 $\pm$ 0.000	10.93 $\pm$ 0.05	0.72 $\pm$ 0.02	1.42 $\pm$ 0.01	1.42 $\pm$ 0.00	0.68 $\pm$ 0.02
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	0.26 $\pm$ 0.08	0.684 $\pm$ 0.001	11.12 $\pm$ 0.06	0.56 $\pm$ 0.02	<u>1.48 $\pm$ 0.01</u>	<u>1.44 $\pm$ 0.01</u>	0.53 $\pm$ 0.02
	A-LQR	<u>0.00 $\pm$ 0.00</u>	0.629 $\pm$ 0.001	24.44 $\pm$ 0.20	0.29 $\pm$ 0.04	1.27 $\pm$ 0.00	1.09 $\pm$ 0.00	0.24 $\pm$ 0.03
	H_∞ (ours)	0.88 $\pm$ 0.09	0.694 $\pm$ 0.001	10.99 $\pm$ 0.02	0.72 $\pm$ 0.05	1.46 $\pm$ 0.01	1.40 $\pm$ 0.01	0.67 $\pm$ 0.05
Llama-3-8B	Original	4.92 $\pm$ 0.71	0.676 $\pm$ 0.001	7.66 $\pm$ 0.03	0.75 $\pm$ 0.02	1.45 $\pm$ 0.01	1.58 $\pm$ 0.01	0.70 $\pm$ 0.03
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	0.84 $\pm$ 0.09	<u>0.685 $\pm$ 0.001</u>	<u>7.85 $\pm$ 0.04</u>	<u>0.66 $\pm$ 0.04</u>	<u>1.50 $\pm$ 0.01</u>	1.62 $\pm$ 0.01	<u>0.65 $\pm$ 0.04</u>
	A-LQR	<u>0.14 $\pm$ 0.07</u>	0.682 $\pm$ 0.001	8.50 $\pm$ 0.03	0.63 $\pm$ 0.03	1.48 $\pm$ 0.02	<u>1.63 $\pm$ 0.01</u>	0.59 $\pm$ 0.03
	H_∞ (ours)	0.14 $\pm$ 0.05	0.648 $\pm$ 0.002	11.01 $\pm$ 0.05	0.44 $\pm$ 0.03	1.48 $\pm$ 0.02	1.25 $\pm$ 0.01	0.40 $\pm$ 0.02
Qwen-2.5-14B	Original	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD
Qwen-2.5-32B	Original	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ITI	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ActAdd	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Mean-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	Linear-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	PID-AcT	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	ODESteer	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	S-PID	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	A-LQR	TBD	TBD	TBD	TBD	TBD	TBD	TBD
	H_∞ (ours)	TBD	TBD	TBD	TBD	TBD	TBD	TBD